

The park visit is spreading out, and thinning out

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1. CLAIM

Between 1979 and 2023, the national park visit became less seasonal and less overnight.

Visits grew about 55%, but almost none of that growth went into the traditional summer camping trip: the June–August share of visits fell from 50% to 41% (–1.6 pp/decade), and overnight stays fell from 118 to 60 per 1,000 visits — a drop that also holds in absolute terms, and inside individual parks, not just in the national mix.

2. DATA AND APPROACH

We use the by-month extract of NPS Visitor Use Statistics distributed by *Responsible Datasets in Context* (Walsh et al., 2024): 33,395 park-months covering all 63 units designated "National Park," 1979–2023, with monthly counts of recreation visits and of overnight stays split into tent, RV, and backcountry.

Because the dataset is a count panel and not a survey of individuals, we cannot observe a visitor. We therefore measured two *ratios*, each designed so that overall growth in visitation cancels out of the numerator and denominator:

Seasonal concentration — the share of a year's recreation visits falling in June, July, and August. This asks when in the year visits happen, independent of how many there are. **Overnight intensity** — overnight stays per 1,000 recreation visits. This asks what kind of visit is happening, independent of how many.

For each ratio we fit an ordinary least squares regression on calendar year, using Newey–West (HAC, 2 lags) standard errors, because annual visitation series are strongly autocorrelated and naive OLS errors would overstate our confidence. Camping-rate trends are fit in logs, so their slopes read as a percent change per decade. Two deliberate choices are worth naming. First, we report each result both visit-weighted (pooling all parks) and equal-weighted (averaging each park's own ratio), because the pooled figure is otherwise driven by a handful of very large parks. Second, we treat 2020–2021 as a potential confound rather than deleting it, and report the trend with and without those years.

3. VISITS ARE DE-SEASONALIZING — BUT BY GROWING THE SHOULDERS, NOT SHRINKING SUMMER

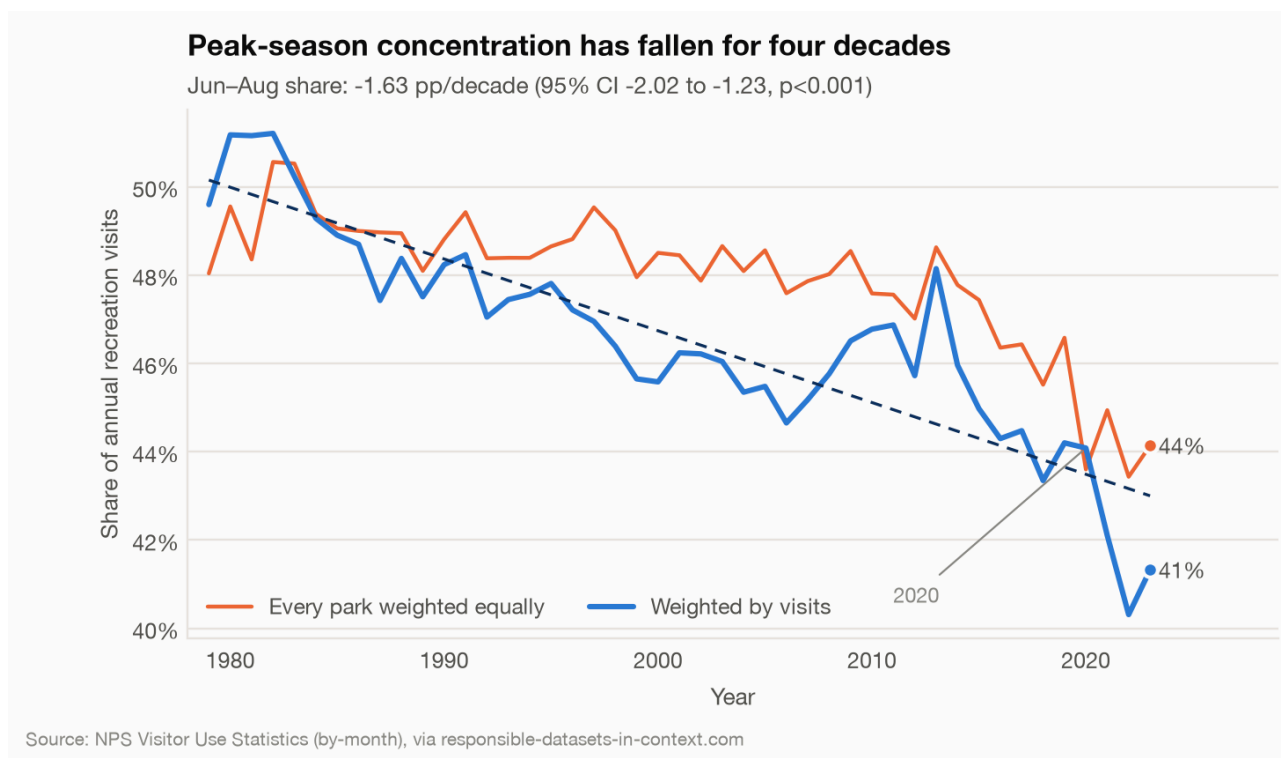


Figure 1. Share of each year's recreation visits occurring in June–August. The blue line pools all parks by visit volume; the orange line averages each park's own share, so small parks count as much as Great Smoky Mountains. Dashed line is the fitted OLS trend on the pooled series. Both decline steadily; 2020 is a visible but not trend-defining excursion.

The June–August share fell from 49.6% in 1979 to a 42.4% average over 2019–2023, a trend of -1.63 percentage points per decade (95% CI -2.02 to -1.23 , $p < 0.001$). Table 1 shows the estimate survives every alternative specification we tried. It attenuates under equal weighting — to -0.96 pp/decade — which tells us the effect is larger in the big parks that most visitors actually go to, but it remains highly significant when every park counts equally. Dropping the pandemic years barely moves it. Defining "peak season" as each park's own best three consecutive months, rather than a fixed calendar summer, also barely moves it, because for the system as a whole June–August is the peak window.

Table 1. Trend in peak-season visit share under four specifications. Slopes are percentage points per decade; HAC(2) standard errors.

Specification	Slope / decade	95% CI	p	Years
Jun–Aug share, visit-weighted (headline)	-1.63	$[-2.02, -1.23]$	1.5×10^{-15}	45
Jun–Aug share, every park weighted equally	-0.96	$[-1.34, -0.58]$	7.7×10^{-7}	45
Jun–Aug share, excluding 2020 and 2021	-1.60	$[-2.04, -1.17]$	3.0×10^{-13}	43
Each park's own best 3 consecutive months	-1.59	$[-1.99, -1.18]$	9.6×10^{-15}	45

A share can fall two ways, and which one it is matters for interpretation. Comparing average annual levels in 1979–1983 against 2019–2023: June–August visits *rose* from 28.3M to 36.6M per year (+29%), while the other nine months rose from 27.6M to 49.9M (+81%). Summer never shrank. De-seasonalization here is entirely a story of the rest of the year growing nearly three times faster. The redistribution is also not confined to the shoulder months one might expect: July (-3.7 pp) and August (-4.4 pp) gave

up share, and the largest single gain went to March (+1.8 pp), followed by December (+1.4), November (+1.1), and October (+1.0).

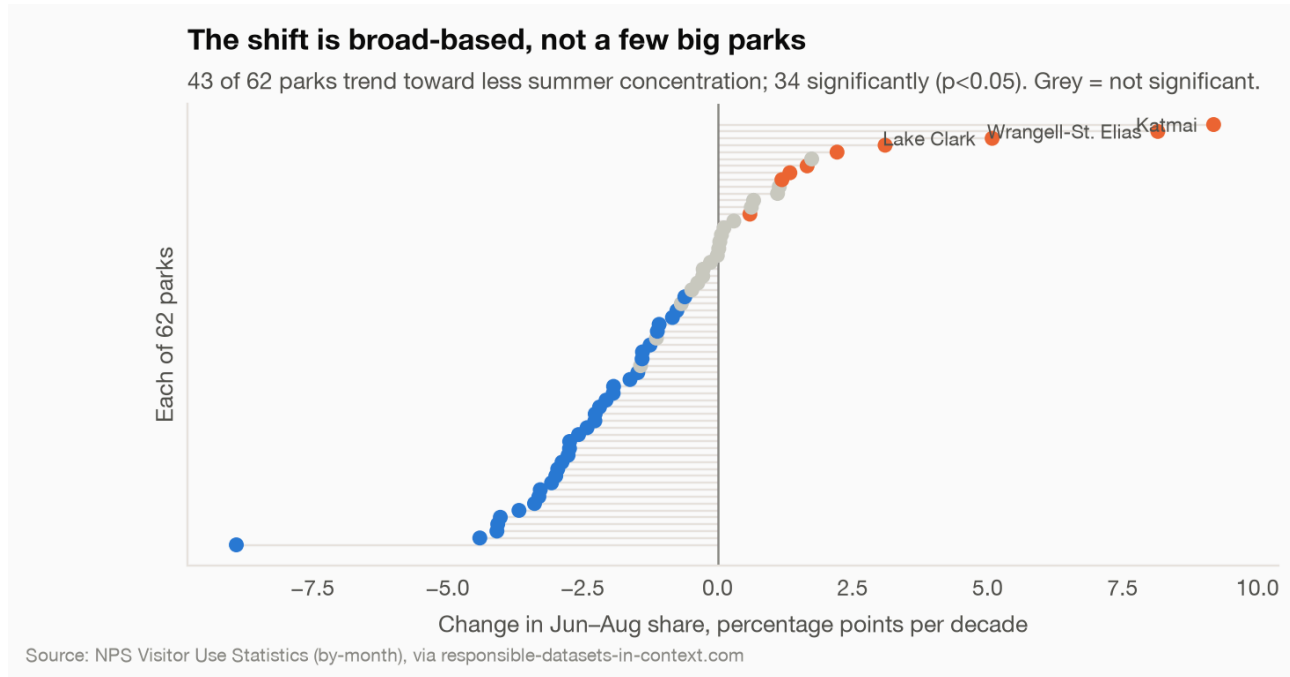


Figure 2. Each park's own trend in June–August share, in percentage points per decade, for the 62 parks with at least 20 years of data. Blue = declining and significant at $p < 0.05$, orange = rising and significant, grey = not significant. The four parks moving most strongly toward summer concentration are all short-season northern parks.

Fitting the same regression separately for each park (Figure 2) confirms this is not an aggregation artifact: 43 of 62 parks trend toward less summer concentration, 34 of them significantly, against only 9 significantly rising. The exceptions are coherent rather than random noise, and they are the most interesting part of the picture. The four parks moving hardest *toward* summer concentration are Katmai (+9.2 pp/decade), Wrangell–St. Elias (+8.2), Lake Clark (+5.1), and Voyageurs (+3.1) — three Alaska parks and one on the Minnesota–Ontario border. These are places where the off-season is not merely unpleasant but largely inaccessible, so growth in demand has nowhere to go except into the short open season. De-seasonalization, in other words, appears to require that a park have a viable off-season to spread into.

4. AND THEY ARE BECOMING LESS OVERNIGHT

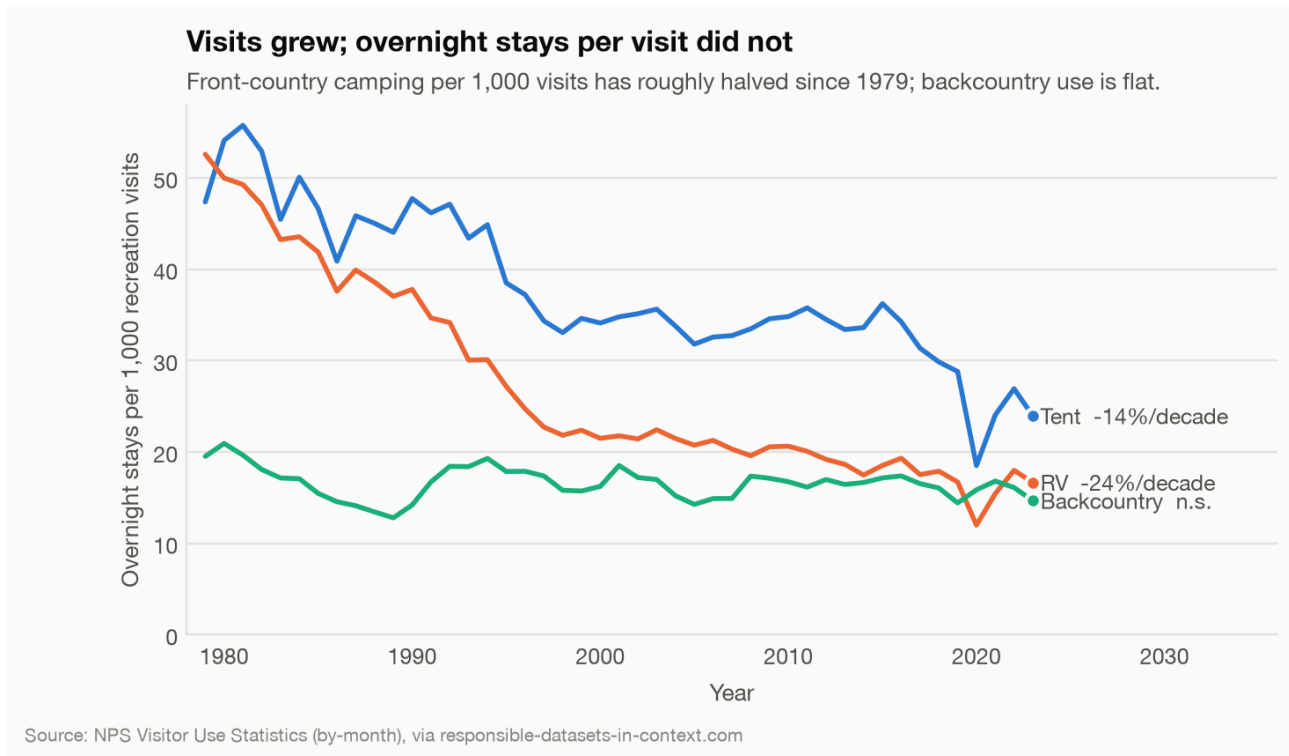


Figure 3. Overnight stays per 1,000 recreation visits, by shelter type. Labels give the fitted trend per decade from a log-linear model. RV camping falls fastest and most steadily; backcountry use shows no significant trend.

Over the same period, overnight stays per 1,000 recreation visits fell from 118 to 60. The decline is concentrated in front-country camping: tent camping fell from 47.4 to 23.9 per 1,000 visits (-14.3% per decade, 95% CI -17.1 to -11.4), and RV camping from 52.6 to 16.6 (-24.0% per decade, CI -26.5 to -21.3). Both are significant at $p < 0.001$. Backcountry use is the exception: it drifted from 19.5 to 14.7 per 1,000 visits, a trend of -1.6% per decade that is not statistically distinguishable from flat ($p = 0.37$). Whatever is displacing the camping trip is not displacing the permit-based wilderness trip.

Two checks rule out the obvious alternative explanations. First, this is not merely a denominator effect from growing day-use: overnight stays fell in absolute terms too, from 6.6M to 4.9M per year (-27%) between 1979–1983 and 2019–2023, while total visits rose 55%. Second, it is not a compositional shift in which growth simply moved toward parks that never had campgrounds. A shift-share decomposition of the national camping rate between 1979–1983 and 2017–2023 attributes the change almost entirely to a within-park term: parks became less camping-intensive, and the change in which parks get the visits contributes roughly -6% (that is, the mix shift slightly *offset* the decline). Tracking individual long-running parks tells the same story — Yellowstone -43% /decade, Zion -23% , Yosemite -19% , all $p < 0.001$ — with one honest exception: Grand Canyon's overnight rate shows no significant trend ($+6.0\%$ /decade, $p = 0.14$).

5. DISCUSSION

What is genuinely surprising. We expected de-seasonalization to look like congestion relief — summer crowds pushed into May and September. It does not. Summer visitation grew substantially in absolute terms; the redistribution went disproportionately into the cold months, with March gaining more share than any other month. That pattern is hard to square with a pure crowd-avoidance story and easier to

square with parks becoming a year-round destination for a larger, differently-distributed population of visitors. The Alaska counter-examples sharpen this: de-seasonalization is not a universal preference shift but something that happens where the calendar permits it.

What is a useful sanity check rather than news. That day use dominates camping is well known. The contribution here is the magnitude and the robustness — a halving of overnight intensity, concentrated in front-country camping, holding within parks — rather than the direction.

What complicates the claim. Three things. (i) Counting methodology has changed repeatedly over 45 years, and the data essay is explicit that counts are modeled estimates from traffic and entrance data, not censuses (Walsh et al., 2024). A slow drift in how off-season traffic is counted would look exactly like de-seasonalization, and we cannot rule this out from within the dataset. (ii) Campground capacity is roughly fixed while visits grew 55%, so a falling camping *rate* is partly mechanical — demand cannot exceed sites. The absolute decline in overnight stays argues against this being the whole story, but not against it mattering. (iii) "Recreation visits" excludes commuters, researchers, staff, and Indigenous people living on park lands, so our claim is about counted recreational visitation specifically.

Generalizability. The seasonal trend is estimated over 45 years and survives removing the pandemic, so we would expect it to continue in the near term — though the 2019–2023 pooled series sits below its own fitted line, suggesting either overshoot or acceleration that a linear model does not capture. We would *not* extrapolate it to short-season parks, where the data show the opposite sign. The camping result is likely to extend to other NPS unit types with campgrounds, but the dataset covers only the 63 units designated "National Park," so this is a conjecture rather than a finding. The specific finding we would most want to test elsewhere is whether the cold-month growth we see reflects changing visitor composition or milder shoulder seasons; separating those requires climate and visitor-origin data that this dataset does not contain.

REFERENCES

- Walsh, M., Jo, E. S., Ramirez, S., Sherman, S., & Tang, E. (2024). *Visits to American National Parks: A Responsible Dataset in Context*. Responsible Datasets in Context. <https://www.responsible-datasets-in-context.com/posts/np-data/> — used for the by-month data extract and, throughout Sections 2 and 5, for how counts are produced and what they exclude.
- National Park Service. *Visitor Use Statistics*. <https://irma.nps.gov/Stats/> — original source of the counts redistributed above.
- Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708. — standard errors used for every trend reported here.
- Analysis code: `src/npdata.py` (statistics), `src/report.py` (every figure and number in this report). Tools: Python, pandas, statsmodels, matplotlib.

APPENDIX A. SHIFT-SHARE DECOMPOSITION OF THE CAMPING RATE

The national camping rate is a visit-weighted average of park-level rates, $R = \sum_p w_p r_p$, where w is a park's share of visits and r its overnight stays per visit. The change between two periods decomposes into a *within* term (parks changed), a *between* term (visits moved between parks of differing camping intensity), and an interaction.

Table A1. Decomposition of the change in overnight stays per 1,000 visits, 1979–1983 to 2017–2023. The symmetric split divides the interaction term evenly between the within and between components, which avoids reporting a large uninterpretable residual.

Component	Raw split	Symmetric split
Rate, 1979–1983 → 2017–2023 (per 1,000)		118.4 → 59.5
Within-park change	83.5%	105.9%
Between-park (visit mix)	–28.3%	–5.9%
Interaction	44.7%	—

Under either split the conclusion is the same: the decline is a within-park phenomenon, and the shift in which parks receive visits worked slightly *against* it. We report the symmetric split in the main text because the raw interaction term here is large — the parks that grew fastest in visit share are also the ones whose camping rates changed most, so the two terms are not cleanly separable.

APPENDIX B. DATA HANDLING AND KNOWN QUALITY ISSUES

We applied only light cleaning: stripping a UTF-8 byte-order mark and trailing whitespace from text fields (several region values arrive as, e.g., "Northeast "), and constructing a month-start date. No visit counts were imputed or adjusted.

Per-park trend estimates (Figure 2) are restricted to parks with at least 20 annual observations, which drops one park and leaves 62. Months reporting exactly zero recreation visits are flagged in the loader; they are genuine non-reporting in some cases (Kobuk Valley has a 2014–2015 gap) and genuine in others (some Alaska parks legitimately record no winter visits), so we did not exclude them from the headline estimates. Carlsbad Caverns is known to have had a malfunctioning counter since 2019. None of these affect the direction or significance of any result reported above: dropping all 1,308 park-months belonging to a park-year that contains a zero-visit month moves the headline seasonal trend from -1.63 to -1.69 pp/decade, well inside the confidence interval in Table 1.